

Continuous Random Variables

6.1 Probability density functions

Definition of Probability Density Function (PDF)

A.K.A density function, a continuous version of the PMF, and this function should always output a **non-negative real value**

The probability is $P(X \in A) = \int_A f(x)dx$

(a) $F(t) = \int_{-\infty}^t f(x)dx$

(b) $\int_{-\infty}^{\infty} f(x)dx = 1$

(c) If f is continuous, then $F'(x) = f(x)$, 即使 f 不連續, 在 f 連續的部分 $F'(x) = f(x)$

(d) For real number $a \leq b$, $P(a \leq X \leq b) = \int_a^b f(t)dt$

(e) $P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b) = \int_a^b f(t)dt$

表示在連續分布中 $P(x) = 0$



Probability density functions

Ex 6.1 Experience has shown that while walking in a certain park, the time X , in minutes, between seeing 2 people smoking has a density function of the form

$$f(x) = \begin{cases} \lambda x e^{-x} & x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

(a) Calculate the value of λ .
(b) Find the cumulative distribution function of X .
(c) What is the probability that Jeff, who has just seen a person smoking, will see another person smoking in 2 to 5 minutes? In at least 7 minutes?

p6.



Probability density functions

Sol:

$$(a) 1 = \int_{-\infty}^{\infty} f(x)dx = \int_0^{\infty} \lambda x e^{-x} dx = \lambda \int_0^{\infty} x e^{-x} dx.$$

By integration by parts, $\int x e^{-x} dx = -(x+1)e^{-x}$.

Thus $\lambda [-(x+1)e^{-x}]_0^{\infty} = 1$.

By l'Hopital's rule we get $\lim_{x \rightarrow \infty} (x+1)e^{-x} = \lim_{x \rightarrow \infty} \frac{x+1}{e^x} = \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0$.

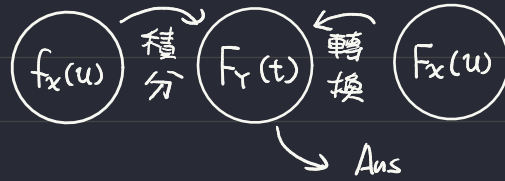
Therefore $\lambda [0 - (-1)] = 1$ or $\lambda = 1$.

p7.

6.2 Density function of a function of a random variable

TLDR: 給定一個隨機變數 X 與一個轉換函數 $Y=h(X)$, 問你 $F(Y)$, $f(Y)$

題目題秒:



Ex 6.3 Let X be a continuous random variable with the probability density function

$$f(x) = \begin{cases} 2/x^2 & \text{if } 1 < x < 2 \\ 0 & \text{elsewhere.} \end{cases}$$

Find the distribution and density functions of $Y=X^2$.

$$1. F_X(x) = \int_1^x 2x^{-2} dx = (-2x^{-1}) \Big|_1^x = -2x^{-1} + 2$$

$$2. F_Y(t) = P(Y \leq t)$$

$$= P(X^2 \leq t) \rightarrow \text{找 } X^2 \leq t \text{ 的上下限}$$

$$= P(-\sqrt{t} \leq X \leq \sqrt{t}) \quad \downarrow \text{注意原始函數範圍}$$

$$= P(1 \leq X \leq \sqrt{t})$$

$$= F_X(\sqrt{t}) = -2t^{-\frac{1}{2}} + 2$$

$$1 \leq t \leq 4 \leftarrow \text{找 } Y=X^2 \text{ 上下限}$$

$$3. f_Y(t) = F_X'(\sqrt{t}) = t^{-\frac{3}{2}} \quad 1 \leq t \leq 4, \quad 0 \text{ otherwise.}$$

6.3 Expectations and variances

Expected value of Continuous R. V. Definition

A continuous R.V. with P.D.F $f(x)$, has the mean/expected value $E(X)$:

$$E(X) = \int_{-\infty}^{\infty} x \times f(x) dx$$

TODO: p.28

Cauchy random variable

這邊只是證明積分後機率加總為 1，可以不用被證明過程

Theorem 6.2

For any continuous R.V. X with distribution function F and density function f ,

$$E(X) = \int_0^{\infty} [1 - F(t)] dt - \int_0^{\infty} F(-t) dt$$

$$E(X) = \int_0^{\infty} P(X > t) dt - \int_0^{\infty} P(X \leq -t) dt$$

TL;DR: 在只有 $F(t)$ 的情況下求 $E(X)$

也可直接將 $F(X)$ 微分得到 $f(x)$

後直接對其作積分 (Def 1)

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} xf(x) dx = \int_{-\infty}^0 xf(x) dx + \int_0^{\infty} xf(x) dx \\ &= -\int_{-\infty}^0 \left(\int_0^{-x} dt \right) f(x) dx + \int_0^{\infty} \left(\int_0^x dt \right) f(x) dx \\ &= -\int_0^{\infty} \left(\int_{-t}^0 f(x) dx \right) dt + \int_0^{\infty} \left(\int_t^{\infty} f(x) dx \right) dt, \end{aligned}$$

where the last equality is obtained by changing the order of integration.

Theorem 6.3

Let X be a continuous R.V. with density function $f(x)$;

then for any function $h(x): \mathbb{R} \rightarrow \mathbb{R}$

$$E[h(x)] = \int_{-\infty}^{\infty} h(x) \cdot f(x) dx$$

且這邊的期望值也遵守線性三條

Definition of Variance

$$\text{Var}(x) = E(X^2) - (E(X))^2$$